

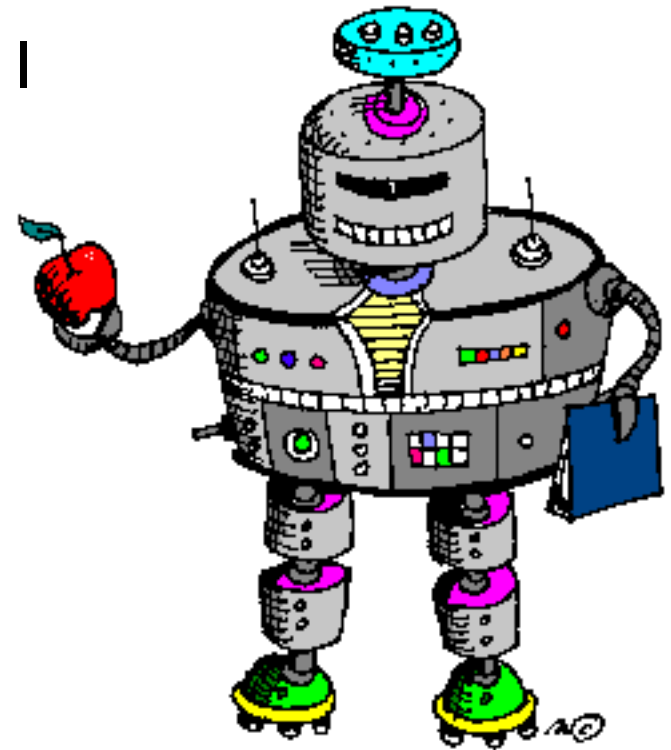
Thinking it Over - Vehicle One

- If the vehicle is in a room of spotlights, it will continue moving, never pausing.
- Is it restless? Does it not like light?
 - It is rather stupid since it can not go back to the dark spots.



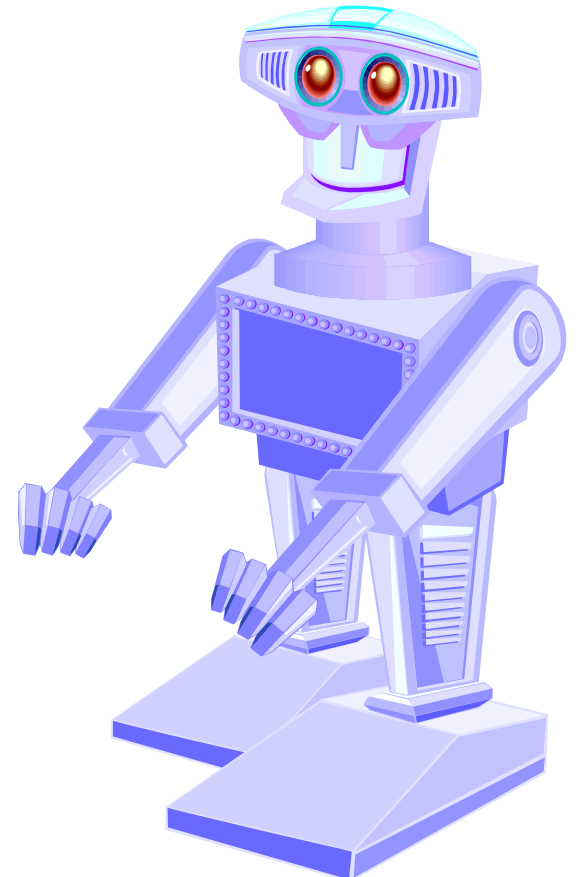
Thinking it Over - Vehicle Two

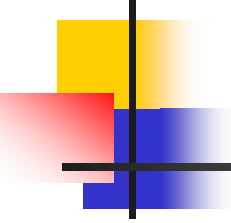
- Do Vehicles IIa and IIb **like** or **dislike** light?
 - IIa) Is restless like Vehicle I and avoids sources, until it finally reaches a place where light is hardly felt.
 - IIb) Unlike IIa, it turns toward light and approaches it quickly.
- How could you describe IIa?
 - **cowardly**
- IIb?
 - **aggressive**



Thinking it Over - Vehicle Three

- Do Vehicles IIIa and IIIb **like** or **dislike** light?
 - IIIa) Loves the light. It will seek bright light and stop there.
 - IIIb) Although it likes light, it explores and searches for even brighter light.
- How could you describe IIIa's feelings towards light?
 - **love**
- How about IIIb's "personality"?
 - **explorative**

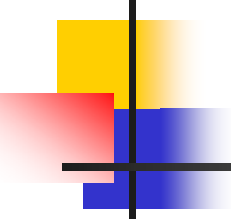




Solution - Vehicle One

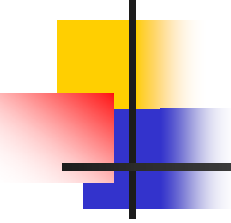
```
/*vehicle1.c*/
#include "logo.h"
#include <dsensor.h>

int main(int argc, char *argv[]) {
    int speed;
    sensorOn(2); /* turn on light sensor attached to 2 */
    while(1) {
        /* the difference of current light and light in the room */
        speed = LIGHT_2 - 25;
        setSpeed(speed); /* sets speed to that difference */
        forward(1);
    }
    return 0;
}
```



Solution - Vehicle Two

```
/*vehicle2.c*/
... /* left out the #include lines */
int main(int argc, char *argv[]) {
    int left, right;
    sensorOn(1);          /* sensors attached to 1 and 3 */
    sensorOn(3);          /* turn them on */
    while(1) {
        left = LIGHT_1 - 25;    /* see vehicle1 solution */
        right = LIGHT_3 - 25;
        leftSpeed(left); /* left speed and right speed */
        rightSpeed(right);      /* are set separately */
        forward(1);
    }
    return 0;
}
```



Solution - Vehicle Three

```
/*vehicle3.c*/  
... /* left out #include lines again */  
int main(int argc, char *argv[]) {  
    int left, right;  
    sensorOn(1);  
    sensorOn(3);  
    while(1) {  
        /* This time the speed is subtracted from 40 so that  
        higher light values give lower speeds. */  
        left = 40 - LIGHT_1;  
        right = 40 - LIGHT_3;  
        leftSpeed(left);  
        rightSpeed(right);  
        forward(1);  
    }  
    return 0;  
}
```